



May 19, 2026

To Whom It May Concern:

I am pleased to recommend Mr. ZHAO Xuecen Jack, a Bachelor of Engineering student in Mechanical Engineering at The Hong Kong Polytechnic University. I served as the Chief Supervisor for his inter-departmental Final Year Project, titled Design and Development of an Electric Toy Car Powered by a Fuel Cell Stack.

Jack contributed significantly to the development of Vandi, an ambitious autonomous toy vehicle powered by a passive vanadium e-fuel cell stack and supported by a hybrid battery power system. The project integrated fuel-cell technology, electrical power management, ROS-based control, LiDAR, camera vision, AI-assisted perception, and a browser-based user interface into one working prototype. In my view, the project was highly innovative and generally impressive, especially given the limited project timeline and relatively small team.

Jack's responsibilities included fuel-cell stack geometry, manufacturing coordination, vehicle integration, ROS/control software, AI/vision implementation, and frontend interaction. He also contributed to additional engineering work such as SolidWorks modelling and simulation, which supported the development and verification of mechanical design. His work showed strong awareness of Design for Manufacturing, Design for Assembly, cross-disciplinary collaboration, and practical system integration. What stood out most was Jack's ability to connect different engineering domains into a coherent final product. The Vandi project required coordination between mechanical engineering, electrical engineering, biomedical engineering, energy conversion, software, and AI. Jack demonstrated the problem-solving skills needed to work across these boundaries and to handle the practical difficulties that arise in real engineering tasks.

Despite challenges such as fuel-cell packaging, electrolyte handling, passive power limitations, manufacturing constraints, and integration risks, Jack remained persistent, resourceful, and technically curious. He approached problems with maturity and was able to refine the design through testing, coordination, and iterative improvement. Based on his performance, I believe Jack has the technical foundation, creativity, and problem-solving ability to handle most engineering tasks with confidence. I am very happy to recommend him for future academic, research, or professional opportunities. If further elaboration on his background and qualifications is required, please do not hesitate to contact me.

Yours faithfully,

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